National University of Computer and Emerging Sciences

BS CS **MT1003 Calculus and Analytical Geometry**

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Quiz : 3.10

Q 1: In the context of a particle moving within the xy-plane, the particle's position is defined by differentiable functions of time, with the rate of change of its x-coordinate, dx/dt, equal to -1 m/sec, and the rate of change of its y-coordinate, dy/dt, equal to -5 m/sec. At the specific point (5, 12) as the particle passes through, what is the rate of change of the particle's distance from the origin? **(5)**

Answer:

The distance from the origin is $s = \sqrt{x^2 + y^2}$ and we wish to find

$$\left.\frac{ds}{dt}\right|_{(5,12)} = \frac{1}{2}(x^2 + y^2)^{-1/2}\left.\left(2x\frac{dx}{dt} + 2y\frac{dy}{dt}\right)\right|_{(5,12)} = \frac{(5)(-1)+(12)(-5)}{\sqrt{25+144}} - 5\text{m/sec}$$

Q2 : In the context of a hydrodynamic system, envision a sophisticated engineering setup in which a continuous efflux of water is taking place from a precisely engineered shallow, inverted conical reservoir. This reservoir boasts a meticulously designed base with a radius of 45 meters and a height of 6 meters. The water is flowing out steadily at an unvarying rate of 50 cubic meters per minute, all while considering the intricate principles of fluid dynamics and geometrical dynamics.

a. At an arbitrary but specific instant, when the water depth stands at 5 meters within the reservoir, calculate with utmost precision the infinitesimal rate at which the water level is receding, expressed in centimeters per minute.

b. Simultaneously, during the exact moment outlined in part (a), determine the infinitesimal rate of change in the radius of the water's surface, considering the minutest changes in time, and express this in centimeters per minute. **(5)**

Answer:

(a) $V = \frac{1}{3}\pi r^2 h$ and $r = \frac{15h}{2} \Rightarrow V = \frac{1}{3}\pi\left(\frac{15h}{2}\right)^2 h = \frac{75\pi h^3}{4} \Rightarrow \frac{dV}{dt} = \frac{225\pi h^2}{4}\frac{dh}{dt} \Rightarrow \left.\frac{dh}{dt}\right|_{h=5} = \frac{4(-50)}{225\pi(5)^2}$

$= \frac{-8}{225\pi} \approx -0.0113 \text{ m/min} = -1.13 \text{ cm/min}$

(b) $r = \frac{15h}{2} \Rightarrow \frac{dr}{dt} = \frac{15}{2}\frac{dh}{dt} \Rightarrow \left.\frac{dr}{dt}\right|_{h=5} = \left(\frac{15}{2}\right)\left(\frac{-8}{225\pi}\right) = \frac{-4}{15\pi} \approx -0.0849 \text{ m/sec} = -8.49 \text{ cm/sec}$

"CHEATING DOESN'T HELP YOU LEARN, AND IT'S YOU WHO MISSES OUT IN THE END."